



**NATIONAL INSTITUTE OF TECHNOLOGY,
KURUKSHETRA-136119**
B. Tech. (RA) – 4th Sem.; Mid-SEM Exam –I
February - 2026

RAPC-212**Time: 50 Minutes****Manufacturing Processes**
Max. Marks: 15

Note: Attempt all questions. Assume suitable missing data, if any.

Q-1: Classify engineering materials and briefly explain the major categories such as ceramics, polymers, and composites. (2)

Q-2: With the help of a neat and properly labeled diagram, explain the stress-strain curve for structural steel, indicating its important regions and critical points. Also discuss the properties of elasticity, plasticity, resilience, proof resilience, and toughness. (3)

Q-3: Classify crystal imperfections in solids. With neat diagrams, explain any point defect, describing its formation and effect on mechanical properties. (3)

Q-4: A strip of steel having an initial thickness of 20 mm is reduced to a final thickness of 15 mm in a single pass rolling operation. The width of the strip remains constant at 200 mm. The average flow stress of the material during rolling is 250 MPa. The roll radius is 300 mm and the rolls rotate at 100 rpm. Calculate, (a) Percentage reduction in thickness, (b) True strain developed during rolling, and (c) Rolling Force. (4)

Q-5 What do you understand by the grade and structure in grinding wheel signature? A newly purchased grinding wheel is coded as; 81 C 80 C 7 R 18. Detail out the designation of the wheel. (3)

End of Question Paper